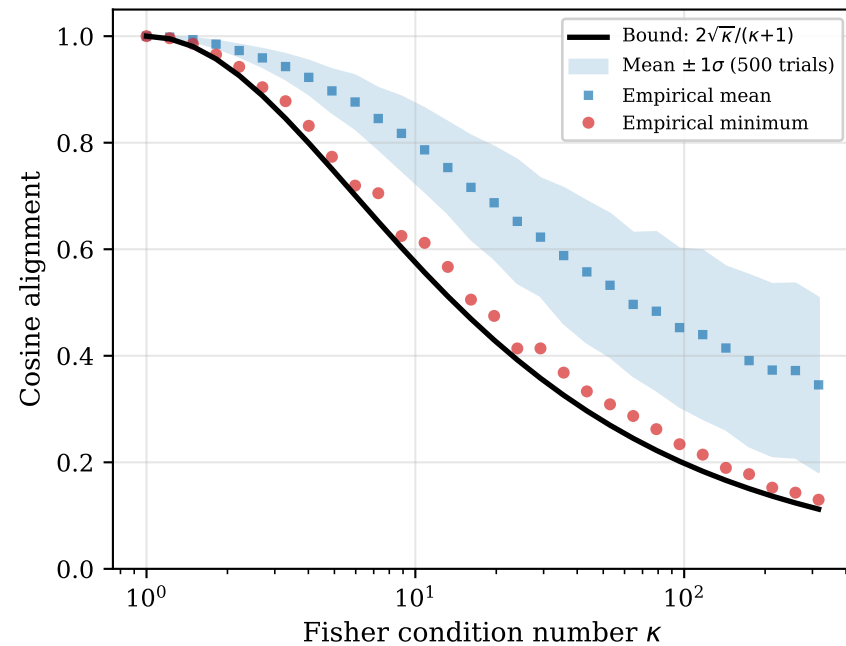
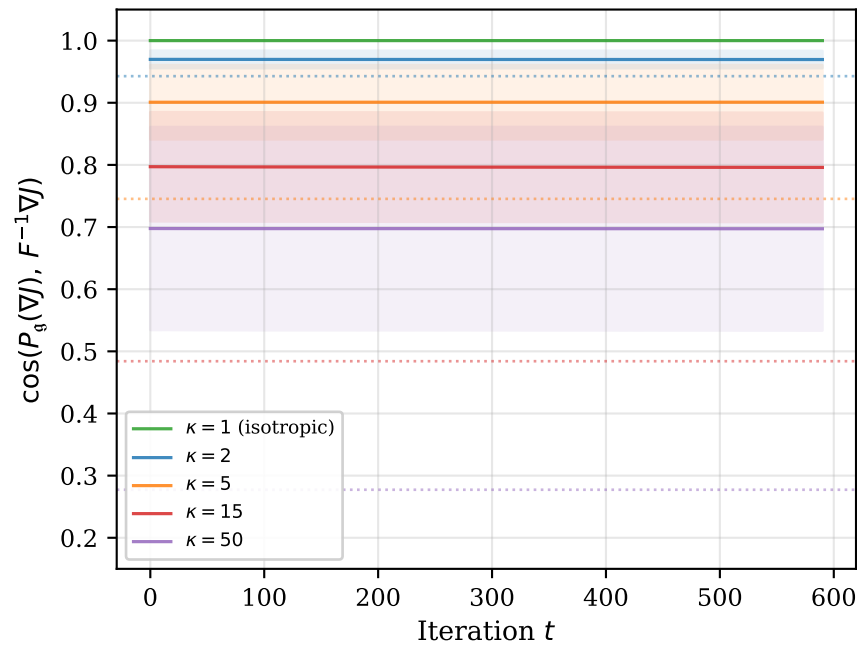


(a) Alignment bound tightness

(b) Alignment during optimization on $\mathfrak{so}(3)$ 

(c) Smoothness dichotomy: compact vs. non-compact

